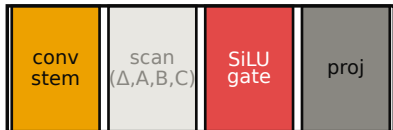


Step 1 -- Freeze the scan (necessity test)

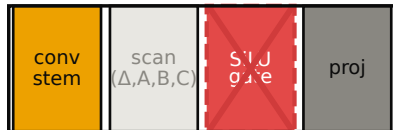
conv stem scan (Δ, A, B, C) SiLU gate proj

BM3-frozen



Step 2 -- Remove gate (localise)

frozen – gate



Step 3 -- Add gate to S4D (sufficiency test)

S4D + gate
+ added



$\Delta(\text{BM3-frozen}) \leq 5\text{pp}$ (CI incl. 0);
vs S4D: +14–20pp, full freeze +21–24pp
→ no scan term is primary

gate removal: –9 to –24pp,
to $\approx$ S4D level or below
→ gate is necessary here

R: narrow 0.17–0.20, wide 0.30
same-size additive control 0.41
→ recovery not gate-specific

Conditional on this block, testbed and budget: the gate's benefit depends on the host.